

Introductory Agentic AI Technical Assignment

Objective

Build a simple **Agentic AI Energy Assistant** that can understand user requests and perform tasks using the provided energy consumption data.

Provided

- One month of simulated hourly energy consumption data in CSV format.
 - timestamp – the start time of the corresponding one-hour interval.
 - energy_consumption_kwh – the amount of energy consumed during that one-hour interval.
 - For example, a reading at 13:00 represents the energy consumed between 13:00 and 14:00.
- Electricity tariff: **SGD 0.25/kWh**

Tasks

The assistant should support the following:

1. Total Energy Consumption

Calculate and return the total energy consumption for a user-specified date range.

Example:

“What was the total energy consumption from 8 August to 14 August?”

2. Energy Invoice Generation

Generate an energy invoice for a user-specified date range.

Example:

“Generate an invoice from 15 August to 21 August.”

The invoice should include:

- Billing period
- Total energy consumption (kWh)
- Energy tariff (SGD/kWh)
- Total amount payable (SGD)

The invoice should be exported as a simple **Excel (.xlsx) file**.

Technical Requirements

- Use Python.
- You may use any free or free-tier LLM of your choice. A paid LLM or paid subscription **is not required** to complete this assignment.
- The LLM should be used to understand the user's request and determine the appropriate available tool(s)/function(s) to complete the task.
- Energy consumption and invoice calculations should be performed using Python code rather than relying on the LLM for calculations.
- The assistant should understand reasonable variations of the example requests rather than relying only on predefined keyword matching.
- The assistant should handle invalid requests or unavailable date ranges appropriately and should not generate or assume missing data.
- A command-line implementation is sufficient. No UI is required.
- Keep the solution simple; a production-level application is not expected.

Submission

Please submit:

- The **source code** for the completed solution.
- A short **screen recording** demonstrating the working solution.
- A brief **written explanation of the approach and how the solution works (maximum 700 words)**.

Before submission, please ensure that **all sensitive information and credentials are removed from the source code and other submitted materials**, including API keys, passwords, access tokens, secrets, or other private credentials.